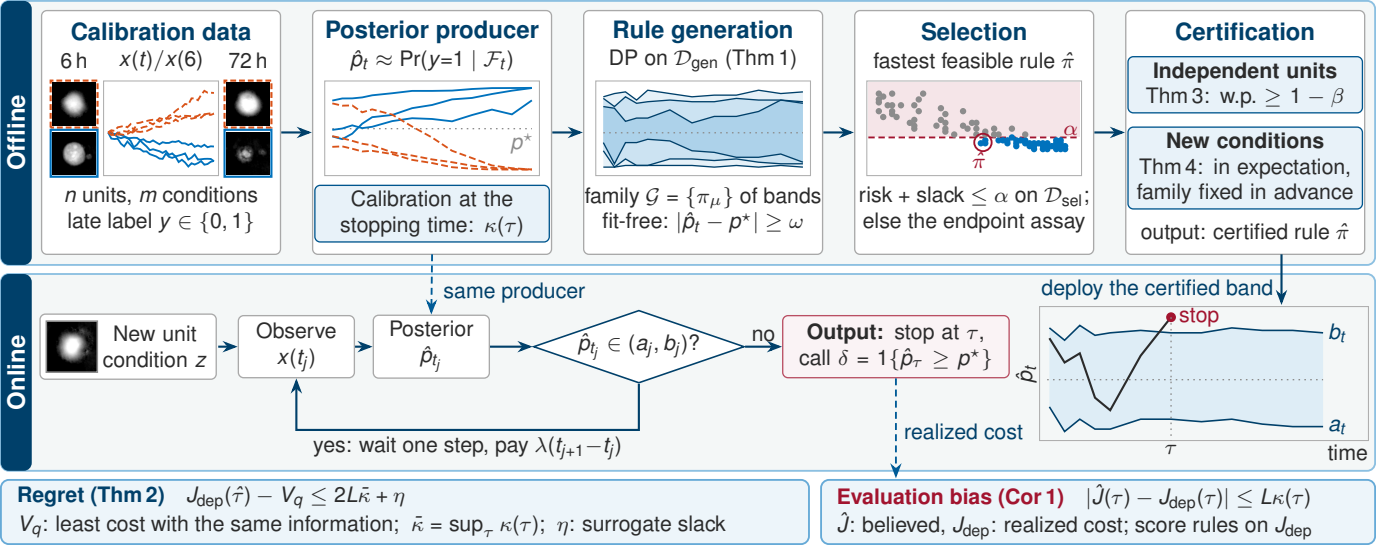

Offline
Calibration data
6 h
x(t)/x(6)
72 h
n units, m conditions
late label y ∈ {0, 1}
Posterior producer
p̂t ≈ Pr(y=1 | ℱt)
p*
Calibration at the stopping time: κ(τ)
Rule generation
DP on 𝒟gen (Thm 1)
family 𝒢 = {πμ} of bands
fit-free: |p̂t − p*| ≥ ω
Selection
fastest feasible rule π̂
α
π̂
risk + slack ≤ α on 𝒟sel;
else the endpoint assay
Certification
Independent units
Thm 3: w.p. ≥ 1 − β
New conditions
Thm 4: in expectation, family fixed in advance
output: certified rule π̂
Online
same producer
New unit condition z
Observe x(tj)
Posterior p̂tj
p̂tj ∈ (aj, bj)?
no
yes: wait one step, pay λ(tj+1 − tj)
Output: stop at τ, call δ = 1{p̂τ ≥ p*}
realized cost
deploy the certified band
stop
bt
at
p̂t
τ
time
Regret (Thm 2) Jdep(τ̂) − Vq ≤ 2Lκ̄ + η
Vq: least cost with the same information; κ̄ = supτ κ(τ); η: surrogate slack
Evaluation bias (Cor 1) |Ĵ(τ) − Jdep(τ)| ≤ Lκ(τ)
Ĵ: believed, Jdep: realized cost; score rules on Jdep